Designing a Universal Information Language

Oscar Bender-Stone

March 26, 2026

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Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph
32	20	Space	48	30	0	64	40	@	80	50	P	96	60	`	112	70	p
33	21	!	49	31	1	65	41	A	81	51	Q	97	61	a	113	71	q
34	22	"	50	32	2	66	42	B	82	52	R	98	62	b	114	72	r
35	23	#	51	33	3	67	43	C	83	53	S	99	63	c	115	73	s
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37	25	%	53	35	5	69	45	E	85	55	U	101	65	e	117	75	u
38	26	&	54	36	6	70	46	F	86	56	V	102	66	f	118	76	v
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47	2F	/	63	3F	?	79	4F	O	95	5F	_	111	6F	o			

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Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

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Background

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[1]Ackoff, R. L. (1989). *From Data to Wisdom*. J. Appl. Syst. Anal., 16, 3–9.
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- Information Management: studies how to **store** and **organize** data

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    - Most useful as *thing*
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Problems

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Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]

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  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data
  - Example: Wikipedia[2]
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
  - **Interoperable:** a portable data format
  - Example: US health records[3]

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Existing Solutions

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Existing Solutions

- **Resource Description Language (RDF):**[1]
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - — No built-in logic engine
- **Labeled Property Graphs (LPGs):**[2]
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - — No logic engine. Hacky tooling/scripts
- **Cyc:**[3]
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - — Hard-coded, proprietary rules

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Proposed Solution

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Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”[1]
- **Goals:**

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- **Welkin**: an information language
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- **Goals:**
  - **Universality:** be completely expressive(**done**)
    - *Any* computer program
    - *Any* "reliable" proof
  - **Scalability:** work at any size (**WIP**)
    - Be efficient
    - Support: future optimizations
    - Describe improvements *in* Welkin
  - **Standardization:** be completely *unambiguous*(**WIP**)

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  - **Standardization:** be completely *unambiguous*(**WIP**)
    - Have a formal specification

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General Requirements

1. Express any computer program

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1. Express any computer program
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1. Express any computer program
   - Need effective operations
   - Established notion: **Turing machines**[1]
2. Enable user-defined *meaning*

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- Achieve both via **representations**

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- Centered around *representing* things
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- Track *faithfulness* of representations via **truth**

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Units

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- Can be broken down, combined
  - Written with braces $\{a, b, \ldots, z\}$

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- Representations are units
- Each unit has its own **context**

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Information

- Representations

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Information

- Representations ~ **Truth**

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Information

- Representations ~ **Truth**
- Information

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Information

- Representations ~ **Truth**
- Information ~ **Proofs**

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Information

- Representations ~ **Truth**
- Information ~ **Proofs**
  - Includes simple facts
  - Expands to complex truh systems
- Major result: can express *any* "reliable" proof
  - More details later! (Metatheory)

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Rules

- **Internal Transitivity:**
- **Contextual Lifting:**

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Rules

- **Internal Transitivity:**
- **Contextual Lifting:**
- **Lattice Laws:**
  - Units as sets

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Turing Completeness

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Turing Completeness

- Part of Universality
- **Theorem:**

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  - φ: some mapping between units and programs
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- Later: completeness for **proofs**!

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Encoding

- Characters written in `type-writer font`[1]
- Define US-ASCII: Printable (codes: 32-126) + `EOF` (code: 127)
- **Token:** collection of contiguous characters
  - Simplifies grammar

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[1]Font used: Intel One Mono (https://github.com/intel/intel-one-mono), licensed under OFL 1.1 (https://github.com/intel/intel-one-mono/blob/main/OFL.txt).

US-ASCII

Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph
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Table 1: Printable US-ASCII codes and glyphs.

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Character Classes

Set Name	Characters
PRINTABLE	Listed in Table 1
WHITESPACE	\t | \r | Space
DELIMITER	{ | } | ' | " | . | ,
RESERVED	DELIMITER | * | @
SQ_CHAR	Any PRINTABLE except ' and \
DQ_CHAR	Any PRINTABLE except " and \
ID_CHAR	Any PRINTABLE not in RESERVED or WHITESPACE

Table 2: Key character classes.

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Building Blocks

- **Invertible Syntax Descriptions:** pairs of functions:[1]

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  - **Printer**: converts a data structure into a string
  - Identity: when parser succeeds, is "reversible" with printer.
- Need **maximal munch:**

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- Need **maximal munch:** consume tokens, not characters

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- **Invertible Syntax Descriptions:** pairs of functions:[1]
  - **Parser**: takes (*consumes*) tokens, converts into a data structure[2]
  - **Printer**: converts a data structure into a string
  - Identity: when parser succeeds, is "reversible" with printer.
- Need **maximal munch:** consume tokens, not characters
  - Simplifies grammar

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[1] Rendel & Ostermann (2010). Invertible syntax descriptions. Haskell '10.
[2] For computer scientists: this is an AST.

The Welkin Grammar

```
start := units - lexeme -> EOF,
units := "" | {unit - lexeme -> *{"",
"," - lexeme -> units}},
unit := node - lexeme -> *{
  ":=" - lexeme -> choices,
  *{"", arc - lexeme -> chain}
},
choices := *{"", "|"} - lexeme ->
choices_list,
choices_list := *{"",
  chain - lexeme -> *{"", "|" - lexeme
-> choice_list}
},
chain := *{"", "*"} - seq -> path -
lexeme -> *{"", arc - lexeme ->
chain},
arc := right_arc | other_arc,
right_arc := "-" - seq -> node - seq -
> "->",
other_arc := "<-" - seq -> node - seq
-> *{"-", "->"},
node := *{"", "*"} - seq -> path,
path := *{"", modifiers} - seq ->
trailer - seq -> *{"", graph},
modifiers := *{
  "@",
  "~" - seq -> *{"", "@"}
},
trailer := segment - seq -> *{"", "."
- seq -> trailer},
segment := HANDLE - lexeme -> *{"",
graph},
graph :=  "{" - lexeme -> units -
lexeme -> "}",
HANDLE := ID | STRING,
STRING := SQ_STRING | DQ_STRING,
SQ_STRING := "'" - seq -> *{SQ_CHAR,
"\'"} - seq_many_till -> "'",
DQ_STRING := '"' - seq -> *{DQ_CHAR,
'\"'} - seq_many_till -> '"',
ID := ID_CHAR - seq_many_till ->
*{RESERVED, WHITESPACE}
```

Listing 1: Full Welkin grammar.

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Outline of Unambiguity

- Simple grammar class: LL(1)

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[1]Consult: Aho et al. (2006). *Compilers: Principles, Techniques, and Tools* (2nd ed.).
[2]Edelmann et al. (2020). *Zippy LL(1) parsing with derivatives*. PLDI 2020.

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  - Unambiguous: every valid input produces *one* output

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  - For details, refer to thesis

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